

Evaluating and Equivalent Expressions*Question Bank 3.06*

Show your work. Circle your final answer.

For questions 1 through 5, apply the laws of exponents to evaluate. Show your steps.

1. $\frac{0.25^{-2} \cdot 0.25^5}{0.25^3}$

2. $(-3)^3 \cdot \left(\frac{-5}{3}\right)^{-4}$

3. $(3^3)^{-3}$

4. $\left(\frac{48}{6}\right)^{-2}$

5. $(3^4 \cdot 2^{-4})^{-1}$

Evaluate each expression in questions 6 through 10. Then name the laws you used.

6. $\frac{6^{-2} \cdot 6^5}{6}$

7. $\frac{(-9)^4}{(-9)^0 \cdot (-9)^3}$

8. $5^2 \cdot \left(\frac{1}{4}\right)^{-1}$

9. $\frac{(-2.5)^0 \cdot (-2.5)^1}{(-2.5)^{-2}}$

10. $\left(\frac{7}{5}\right)^{-4}$

Common bases.

11. Write each expression as a single power with the base given.

a. 8^3 with base 2

b. 25^3 with base 5

c. 64^2 with base 8

d. $2^4 \cdot 4^2$ with base 2

12. Marquise says $\left(\frac{1}{8}\right)^3 \cdot 2^{-3}$ and $\left(\frac{1}{4}\right)^5 \cdot 2^{-2}$ are equivalent because both can be written as 2^{-12} .

a. Write the first expression as a single power of 2. Show your work.

b. Write the second expression as a single power of 2. Show your work.

c. Is Marquise correct?

13. Which expression is equivalent to $\frac{2^6}{3^3}$?

☐ A. $3^{-3} \cdot 2^6$

☐ B. $\left(\frac{2}{3}\right)^3$

☐ C. $\left(\frac{2}{3}\right)^2$

☐ D. 6^3

14. Find the value of n that makes each statement true.

a. $8^n = 2^{12}$

b. $\left(\frac{1}{9}\right)^n = 3^6$